

A Frequency Dial for Structure-Conditioned Diffusion

SUBMISSION ID

Abstract

A structure-conditioned diffusion adapter is trained for one operating point. How closely its output follows the condition is fixed at training, and loosening it later means another checkpoint or a lower adapter strength. We move that choice into the condition. A single adapter on Stable Diffusion 1.5 is trained on grayscale images passed through a radial high-pass filter, with the cutoff r drawn per sample. At test time r becomes a dial, much as image-editing tools let users pick a frequency band. On FFHQ-512, raising r from 0.05 to 0.3 lowers structure consistency (mean of three training seeds) from 0.7479 to 0.2318 while FID stays between 51.58 and 54.68. In our samples the face layout holds while background, clothing and colour drift from the reference. Lowering the adapter’s strength instead pays for looser structure with FID: at strength 0.50 structure is looser than at $r=0.3$ under the same fixed instrument, and FID is 76.46 against 52.77. At $r=0.1$, an adapter trained only for that cutoff still follows the structure more closely. Injecting through the UNet skip tensors carries most of the structure, and an artist can filter a hand drawing the same way; the output’s correlation with the drawing falls as r grows.

CCS Concepts

• **Computing methodologies** → **Image manipulation**; **Neural networks**;

1. Introduction

Image editing has long given users a scale or frequency band to turn [FFLS08, PHK11], and sketch-based face synthesis has treated a drawing as a soft constraint rather than a template [CSG*20], in some systems with an explicit control over how far the sketch is trusted [YWL20]. ControlNet [ZRA23] and its successors [MWX*24, ZCC*23, QZY*23, LYK*24] moved that control into training. An adapter trained on Canny edges follows Canny edges; one trained on depth follows depth. Following a structure more loosely then means picking a different checkpoint or scaling the adapter down at inference.

We change what the condition is rather than how hard the model listens to it. Multi-scale editing removes or amplifies a band of the output image [FFLS08, PHK11]; here the band is removed from the condition, and the generative prior decides what fills it. The condition is the image after a radial high-pass filter with normalised cutoff r . At $r=0$ the model receives the grayscale image and the task is colourisation. As r grows the filter removes more of the spectrum, until at $r=1$ almost nothing is left: measured on 500 FFHQ test images, a cutoff of 0.05 already discards 88.6 % of the spectral energy outside the constant term. Training draws r per sample, so one set of weights sees the whole range, and r becomes an inference-time parameter. At a fixed seed, raising r leaves the face layout in place while background detail thins and clothing and colour drift from the reference (Figure 3). Lowering the adapter’s strength is a different control: at low strength pose and framing change, and FID rises (Sec. 4.3).

Our contributions are:

- A frequency dial on the condition. Randomising the high-pass cutoff during training gives one adapter an inference-time control over structure adherence on $r \in [0.05, 0.3]$ at a steady FID band (Sec. 4.2); it trails a per-cutoff specialist at $r=0.1$ and leads one at $r=0.3$ (Sec. 4.4).
- The dial is not adapter strength. Under one fixed instrument, strength settings around the cutoff’s structure consistency sit at higher FID and change pose and framing (Secs. 4.3 and 4.1).
- Where the condition should enter a UNet (Sec. 5). On SD1.5 the skip-tensor path carries most of the structure that cross-attention alone does not (Secs. 5.2 and 5.3), and with attention-only injection a convolutional encoder reads the condition better than the frozen VAE (Table 5).

2. Related work

Conditional control. Text-to-image diffusion [HJA20, SME21, RBL*22, NDR*22, SCS*22], surveyed for visual computing by Po et al. [PYG*24], became spatially controllable with ControlNet [ZRA23], which reads the condition image with a small convolutional network, duplicates the encoder and injects residuals through zero convolutions. T2I-Adapter [MWX*24] does the same job with a small feature extractor, IP-Adapter [YZL*23] adds decoupled cross-attention for image prompts, and GLIGEN [LLW*23] grounds generation on boxes. Uni-ControlNet [ZCC*23] and UniControl [QZY*23] fold many condition types into one model, and ControlNet++ [LYK*24] closes a consistency loop on the produced image. A second line shrinks the control branch on the same UNet: ControlNet-

XS [ZFR24] couples a much smaller control network to encoder and decoder in both directions, CtrlLoRA [XHSC25] adapts a shared base ControlNet to each new condition with LoRA, and Simple-ControlNet [FCS*25] feeds the condition through a single-scale projection at the top of the UNet with LoRA on self-attention, in under 3M parameters, exposing a scale on that projection at inference. Each of these fixes the condition type at training and, with it, how tightly the output follows; an inference-time scale such as Simple-ControlNet’s moves that operating point by weakening the adapter. Our cutoff moves it through the condition on one set of weights, and on our adapter the two controls separate in FID (Sec. 4.3).

Efficient control on transformer backbones. As backbones moved from UNets to diffusion transformers [PX23, EKB*24], control moved from duplicated encoders to lightweight modules. ControlNeXt [PWZ*24] removes the parallel branch, OminiControl [TLY*25] reuses the backbone itself as the condition processor, and EasyControl [ZYS*25] and NanoControl [LZL*25] inject conditions through LoRA-style [HSW*22] modules at a small fraction of the parameter cost. Recent work also revisits how conditions enter these models: Laytrol [HHLZ26] inherits the backbone’s parameters to preserve pretrained knowledge under layout control, PixelControl [LLZ*26] notes that the VAE’s spatial compression can weaken high-frequency and low-area condition signals, and Zhou et al. [ZWC*26] replace concatenate-and-attend with position-aligned and keyword-scoped attention in multi-condition DiTs. On a UNet the question has a concrete form, since skip tensors are available as an entry point: in our comparison (Sec. 5.2) skip residuals carry more structure than cross-attention.

Frequency bands as an editing axis. Band decompositions are a standard editing axis in graphics: detail manipulation rescales the bands of an edge-preserving or Laplacian decomposition [FFLS08, PHK11], and hybrid images combine the low spatial frequencies of one picture with the high frequencies of another, so that the interpretation changes with viewing distance [OTS06]. In diffusion, frequency decomposition appears mostly as an intervention on the model’s own signals rather than on its condition. FreeU [SHJL24] rebalances skip and backbone features spectrally at no training cost, FreSca [HLT*25] scales frequency bands of the noise prediction using spatial or energy-based cutoffs, DeCo [MWW*26] decouples frequencies in pixel-space diffusion, and FeRA [YHZ*26] routes adaptation by frequency energy. FOCUS [TZY*26] conditions the reverse process on learned frequency priors for test-time adaptation. Closest to us in the choice of control, frequency bands also act as a control on image-to-image guidance: FCDiffusion [GXZL24] extracts DCT bands of the source features and switches among frequency control branches, FBSDiff [GL24] sets the guiding intensity of a reference through the bandwidth of a substituted band without training, and ILVR [CKJ*21] conditions a pretrained DDPM on a reference at a chosen downsampling factor without further learning. Filter-guided diffusion [GYD24] applies a spatial filter to the denoised estimate at inference and exposes its scale and the guidance strength as separate controls without training; our band is instead part of what the adapter is trained to read. Prasain [Pra26] conditions on wavelet bands instead of edge maps for map-to-satellite synthesis. None of them randomises the condi-

tion’s *bandwidth* during training so that the band becomes a control at inference. Randomising a conditioning parameter while training and exposing it at inference is not itself new: SDXL [PEL*24] samples crop coordinates during data loading and surfaces them as a user control, and Imagen [SCS*22] draws the augmentation level of its low-resolution condition at random in training and sweeps it at inference, there to find the best sample quality rather than to give the user a control. Our delta is the choice of parameter and its measured separation from adapter strength: bandwidth and adapter strength are different controls (Sec. 4.3).

Drawings as soft constraints. Sketch-based face synthesis has treated rough drawings as soft constraints [CSG*20] and, closest to our goal, lets the user set how far a drawing is trusted through a refinement level [YWL20]; diffusion-based stroke and sketch synthesis likewise lets the user decide the level of faithfulness to the input [CCC*23]. With pretrained diffusion models, sketch guidance can be applied at inference through a small latent guidance predictor instead of a dedicated encoder [VACO23], and the noise added before denoising balances faithfulness to the input against realism without training [MHS*22]. XDoG [WKO12] moves a photograph from pencil shading to hatching and woodcut by changing the parameters of one filter family. Our cutoff is a filter parameter of the same kind, set on the condition and left for the generative prior to complete. Our control differs in what it varies: the information content of the condition, set by a frequency cutoff, on one adapter whose response we measure separately from its strength. Because the condition is a continuum and not a fixed extractor, a single adapter can expose it as a dial. StructDiff [HLL*26] targets structure-preserving, spatially controllable single-image generation, varying samples within one source image’s own statistics while keeping its layout; we hold an external condition instead, and the cutoff sets how much of it the output has to reproduce.

3. Method

Condition. Given an image x , we convert to grayscale, take the 2D FFT, zero every frequency inside a disc of radius $r \cdot \min(H, W)/2$, invert, and min–max normalise to $[0, 1]$. The normalisation is deliberate: it removes the absolute scale of the filter response, so the model has to infer structure density from spatial statistics. We hypothesise that this is what lets a hand drawing, which carries no r , be used at test time. The filter is a high-pass because the point is to keep shape and hand the rest to the prior: edges and texture sit in the high band, tone and colour in the low one, the split two-scale tone and detail editing [BPD06, FFLS08] works with, so a low-pass or band-pass condition would give the model the part we want it to invent. What we add is that the boundary is a parameter rather than a fixed extractor. The mask is an ideal disc rather than a smooth roll-off so that r names exactly which frequencies are gone, at the cost of the ringing visible in Figure 1, which the model is trained on and which we did not measure separately. Figure 1 shows the mask, spectrum and condition at each cutoff.

Condition encoder. A convolutional encoder reads the condition image directly at 512^2 and reduces it $8\times$ to the latent grid: seven 3×3 convolutions with SiLU activations, three of them with stride 2 and followed by group normalisation, widening from 32 to 128

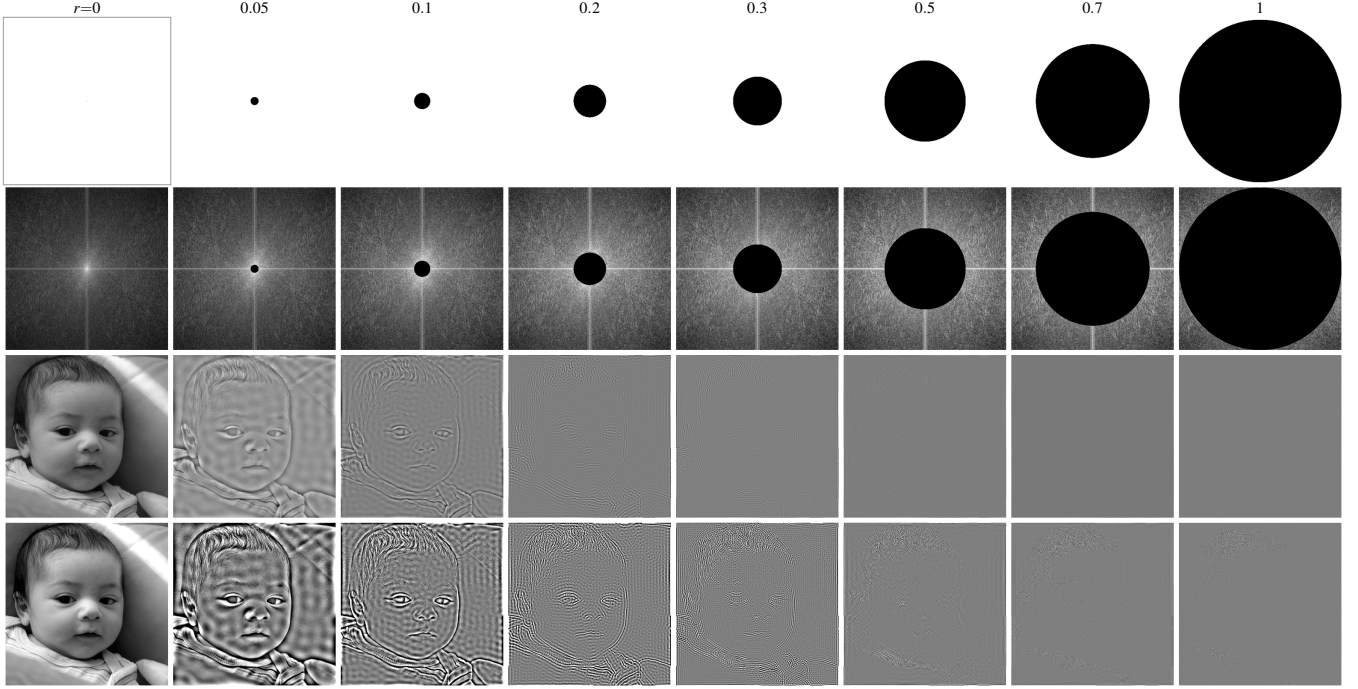


Figure 1: The dial, from the frequency side. Columns are $r = 0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0$. **Row 1:** the radial mask; black is removed. **Row 2:** the log-magnitude spectrum after masking, with the hole growing outward. **Row 3:** the resulting condition, exactly as the model receives it. **Row 4:** the same data displayed at $\pm 2\sigma$ so that the residual content is visible. Above $r \approx 0.2$ the condition looks like flat grey (averaged over 500 test images, its standard deviation falls from 0.238 at $r=0$ to 0.034 at $r=1$), yet the face outline survives in row 4, and the model still reaches SC 0.4019 at $r=0.2$. The per-image normalisation leaves the model only this low-contrast residue to read structure from.

channels before a final projection to 4. Each attention site reads the encoder output through a 1×1 projection. The ablations call an optional 3×3 convolution of 64, 128 or 256 channels inserted before that projection a *stem*; the main model has none. We read the condition with this convolutional encoder rather than the frozen VAE; Sec. 5.1 reports the round-trip measurement.

Injection. Encoder features enter the UNet in two places (Figure 2). Zero-initialised 1×1 heads produce one residual per decoder skip tensor and one for the mid block, added through the UNet’s existing ControlNet interface. In parallel, each cross-attention site attends over a 3×3 window of the condition features, gated by a zero-initialised per-channel vector. At 1×1 the softmax runs over a single position, the site reduces to a gated value projection, and structure consistency is not lower beyond the seed spread (Sec. 5.2). Both paths start as exact no-ops: with them switched off the network reproduces stock Stable Diffusion 1.5 [RBL*22] bit for bit, which we assert before every training run.

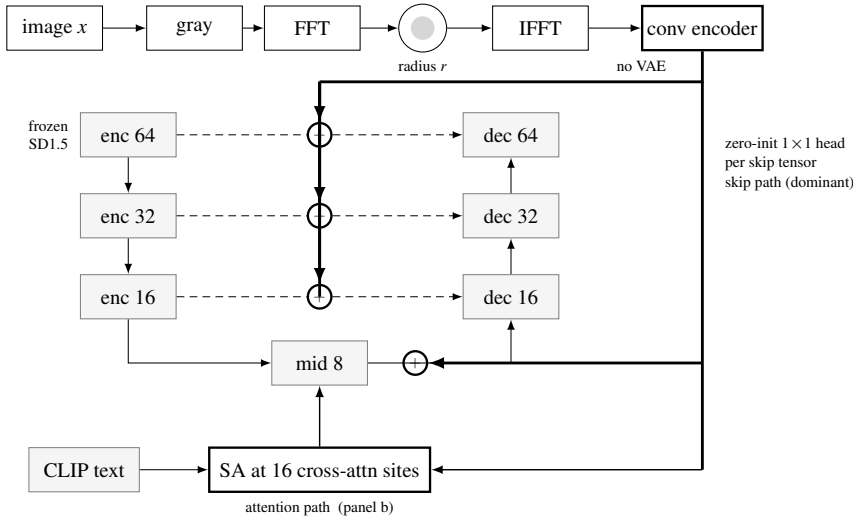
Training. SD1.5 [RBL*22] stays frozen. We train 6,611,469 parameters on FFHQ-512 [KLA19] (images under their Flickr authors’ Creative Commons or public-domain licences, the dataset under CC BY-NC-SA 4.0) with BLIP [LLXH22] captions, split 65,100 train / 1,400 validation / 3,500 test by a seeded shuffle of the 70,000 images (5% test, 2% validation, seed 2026). AdamW,

learning rate 10^{-4} , $\beta = (0.9, 0.999)$, cosine schedule with 500 warm-up steps, gradient clipping at 1.0, batch 32, bf16, 12 epochs on one H200. The loss is ϵ -prediction with Min-SNR weighting [HGL*23], $\gamma = 5$. Text conditioning is dropped with probability 0.1 for classifier-free guidance [HS22] and the structure condition independently with probability 0.05. The condition is grayscale so that the filter reads one channel and a cutoff means the same thing in every image, leaving colour to the prior; Table 6 keeps a colour-condition row as the control for that choice, and a no-dropout row for the structure-condition dropout of 0.05. Training stops at 12 epochs, where the validation curve has flattened. 0.05. AdamW uses the PyTorch default weight decay of 0.01. An EMA of the adapter (decay 0.9999) is kept with a 10-step interval; evaluation uses the raw weights. r is drawn uniformly from $[0, 1]$ per sample, so one adapter sees the whole range; we report $r \in [0.05, 0.3]$ because that is where the control is usable, the sweep being non-monotone at the left end and not separable across seeds at the right (Sec. 4.2). No narrower training range was tried. Table 1 lists the remaining settings.

4. Experiments

Tables numbered with an S prefix are in the anonymous supplementary material.

(a) condition path and the two injection sites



(b) inside one site

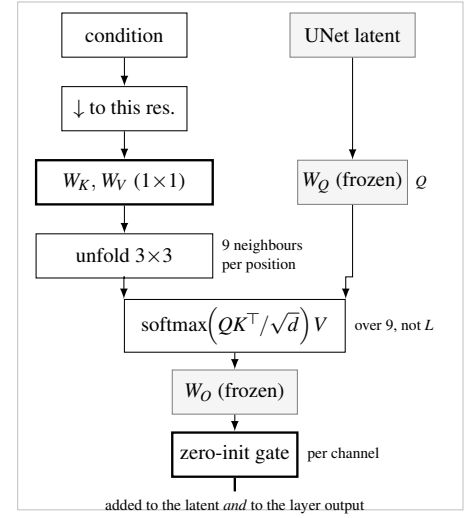


Figure 2: (a) A disc of radius r is masked out of the spectrum and a trainable convolutional encoder reads the result at full resolution, never through the VAE. Its features enter the frozen UNet twice: as zero-initialised residuals on each decoder skip tensor ($\oplus$; three resolution levels and the mid block are drawn, 12 skip heads and one mid head in the network), and through StandAlone attention at the 16 cross-attention sites. Stroke weight follows the measured split: at $r=0$, SC is 0.5370 with only the skip path active and 0.2112 with only attention (Sec. 5.2). (b) One StandAlone site. The query comes from the UNet’s own latent through the frozen W_Q ; keys and values come from the condition through trained 1×1 projections, unfolded so each position attends over its 3×3 neighbourhood rather than the whole map; a 1×1 window performs at least as well (Sec. 5.2). The result passes the frozen output projection and a zero-initialised per-channel gate, then is added both into the latent — so the text query below sees structure — and to what the layer returns.

4.1. Metrics

We evaluate on 500 of the 3,500 held-out images with a DDIM sampler at 50 steps, CFG 7.5, seed 2026.

Structure consistency. SC converts the generated image to grayscale, applies the same high-pass filter at the same cutoff the model was given, and takes the Pearson correlation against the given condition over the flattened pixel vector, per image, averaged over images. Both signals are the per-image min–max normalised filter output, so the correlation is scale free. Two anchors bound it: feeding the ground truth back gives exactly 1.000 at every r , and scoring a generation against the condition of a different test image gives 0.0847 at $r=0$ and 0.0082 at $r=0.1$. Because the instrument moves with r , a row-to-row difference mixes model behaviour with instrument change, so we report a cutoff-independent metric beside it and a fixed-instrument reading below.

Edge-F1. Canny edges (thresholds 100 and 200 on the 8-bit grayscale image) are taken from the generation and from the reference and matched with a 2-pixel tolerance, implemented as a 5×5 dilation of each map before intersecting; F1 is computed per image and averaged. It does not depend on r , which is why it appears in Table 2: 0.7796 at $r=0.05$ falling to 0.4724 at $r=1$, so the dial survives a cutoff-independent instrument.

Diversity. Mean pairwise LPIPS (AlexNet backbone) over $K=4$ samples generated from one condition with different seeds; higher

means the condition constrains less. It is not independent of the LPIPS column (defined below): across the sweep LPIPS – Diversity is 0.0320 ± 0.0038 , nearly constant while Diversity spans 0.21, so Diversity carries close to the same information as LPIPS and serves as a secondary reading.

FID, LPIPS and CLIP. Clean-FID [HRU*17, PZZ22] in its clean mode at 512 pixels, between the 500 generations and the same 500 references. The LPIPS column is AlexNet LPIPS [ZIE*18] between each generation and its own reference, averaged over images. CLIP score is 100 times the cosine similarity of ViT-B/32 [RKH*21] embeddings of the image and its own caption.

Fixed instrument. Structure consistency re-extracts the condition at the same cutoff the model was given, so the measuring instrument moves with the thing it measures. We therefore also re-score the saved generations with the cutoff held at $r_m=0.1$ throughout. This reads the saved 8-bit images, so at $r=0.1$ it gives 0.6210 where Table 2, computed before quantisation, gives 0.6211.

Code and data. Code, configuration files, the per-run result files and the main model’s checkpoints are in an anonymous repository, https://anonymous.4open.science/w/Frequency_Adaptation-3125/, and every number in the paper is generated by one script from those result files.

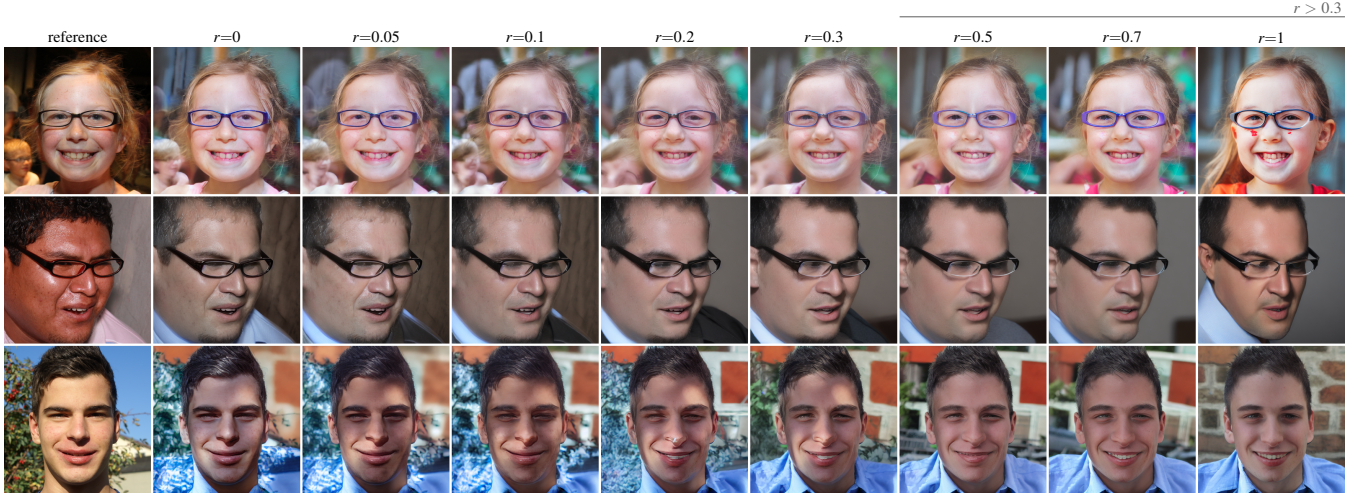


Figure 3: The main model (training seed 2026) with sampling seed 2026 and each reference’s BLIP caption; along a row only r changes. The rows are the first three test images in generation order, and the tiles are the saved generations scored in Table 2. The reference is never shown to the model. As r rises, background detail thins and clothing and colour drift from the reference while the face layout holds; at $r=1$ colour artefacts appear on the first row. All outputs are generated by the model under study.

4.2. The dial

Table 2 gives the operating curve. On $r \in [0.05, 0.3]$ structure consistency falls at every step while FID stays between 51.58 and 54.68, and sample diversity rises alongside. Going from $r=0$ to $r=0.05$ structure consistency rises by 0.0479 ($z = 6.24$ with the unpaired standard error over the 500 images), so the sweep is not monotone at its left end. Edge-F1 moves the same way, from 0.7776 to 0.7796, but we have no uncertainty for that change, so it does not separate model behaviour from an artefact of the SC instrument. We read the rise as the $r=0$ condition, the whole grayscale image, carrying so much of the low-frequency field that it competes with the edges the adapter must reproduce; cutting a small disc out leaves the shape easier to follow. We did not measure this. At the right end $r=0.5$ and $r=0.7$ are not separable across seeds (paired $t = 0.732$, $df = 2$, critical 4.303).

Figure 3 holds the seed and the caption fixed and varies only r . Because the caption is derived from the reference, it could be carrying part of the structure. We repeated the sweep with one generic caption, “a photo of a person”, for every image. Structure consistency stays close (0.7039, 0.6179 and 0.2196 at $r=0$, 0.1 and 0.3, against 0.6991, 0.6211 and 0.2347 with captions) while FID rises to 55.56–65.84, as it should once the caption stops describing the image. The structure comes from the condition. Figure 4 plots the curve and, below it, the cutoff and the two scaled adapters in the structure-consistency/FID plane.

Across seeds 2026, 2027 and 2028 the curve is stable: SC at $r=0.05$ is 0.7479 ± 0.0015 and at $r=0.3$ it is 0.2318 ± 0.0106 .

Re-scored with the fixed instrument of Sec. 4.1, the curve still falls: 0.6524 at $r=0.05$, 0.6210 at $r=0.1$ and 0.2599 at $r=0.3$. Holding the instrument fixed compresses the range, as it must, and keeps the decline.

4.3. Adapter strength as the alternative knob

An obvious alternative is to train on one cutoff and turn the adapter’s contribution down at inference, the knob lightweight control modules expose [FCS*25]. We ran it on the adapter trained at fixed $r=0.1$ (Sec. 4.4), scaling both injection paths, and, as a second check, scaled the randomised- r adapter at the same cutoff (Table 3; Table S5 sweeps it finely).

Both scaled curves leave the FFHQ image distribution as they go. At scale 0.25 the fixed- r adapter reaches FID 102.43 and diversity 0.7689, beyond anything the cutoff produces over its whole sweep, where FID stays within 50.13–57.40 and diversity peaks at 0.5066. Turning the randomised- r adapter down removes structure while FID rises, to 76.46 at scale 0.50 (70.91 for the fixed- r adapter). The fine sweep of Table S5 brackets the cutoff $r=0.3$, which reads 0.2599 at FID 52.77, each scored at $r=0.1$: scale 0.6 gives 0.3373 at FID 66.39, and scale 0.5 gives 0.2146 at FID 76.46. We read the cost as a train–test mismatch rather than a weaker condition: the adapter is only ever trained at scale 1.00, so turning it down puts the UNet in a state it never saw, while r was drawn per sample and moving the cutoff stays inside the training distribution. Figure 5 agrees: at scale 0.25 both adapters still write the grayscale condition through their scaled skip path, too weakly for the model to complete it, and the second row comes out grey under either one. We did not measure this. At full strength the fixed- $r=0.1$ adapter reaches FID 48.51 against 54.68 for the randomised- r adapter at the same cutoff. Figure 5 shows what the scaled settings produce for the first two test images.

4.4. Against per-cutoff specialists

We trained four adapters at fixed r , one run each, and compared each against the randomised- r adapter, trained three times, at its own cutoff. We divide each gap by the randomised- r adapter’s

Item	Setting
Base model	stable-diffusion-v1-5/stable-diffusion-v1-5, frozen
Condition encoder	seven 3×3 convolutions, SiLU, three with stride 2 and group normalisation, 32 to 128 channels, 4 output channels
Trainable params	6,611,469 (main); skip only 6,205,856; attention only 405,613
Data	FFHQ-512, 65,100 / 1,400 / 3,500, seeded shuffle (seed 2026); the validation split only monitors training on four images
Sketch data	a573p/Face-Sketches40, all 40, and rhfeiyang/photo-sketch-pair-500; neither card states a licence
Optimiser	AdamW, lr 10^{-4} , $\beta=(0.9, 0.999)$, weight decay 0.01, cosine, 500 warm-up steps
Batch / precision	32 / bf16, gradient clipping 1.0, 12 epochs, Min-SNR $\gamma=5$, one H200
Compute	5.7 h per training run on one H200 at 39 images/s, peak 61 GB; 1.88 s per sampled image (DDIM 50, CFG 7.5, batch 1, 512 px), peak 2.9 GB; 156 GPU-hours over 26 training runs and about 97 more for the 186,160 images generated, at the measured rate, 253 in all
EMA	decay 0.9999 every 10 steps; evaluation uses the raw weights
Dropout	text 0.1, structure condition 0.05
Training seeds	main model 2026, 2027, 2028; specialists, ablations and skip only 2026
Sampling	DDIM, 50 steps, CFG 7.5, seed 2026; Diversity seeds $2026+10^5+i$ for image i
Evaluation	first 500 of the 3,500 test images; clean-FID, clean mode, 512 px; LPIPS AlexNet against the reference; CLIP ViT-B/32 (OpenAI weights), 100× cosine
Baselines	llyasviel/sd-controlnet-canny, llyasviel/sd-controlnet-hed, Ten-centARC/t2iadapter_canny_sd15v2; Canny 100/200 for all; conditioning scale 1.0; default PNDM sampler, 50 steps, CFG 7.5, seed 2026
Software	Python 3.12.3, PyTorch 2.10 (NGC 25.11), diffusers 0.32.2, clean-fid 0.1.35, lpips 0.1.4, open_clip 2.30.0

Table 1: Settings behind every table, read from the run manifests and evaluation scripts.

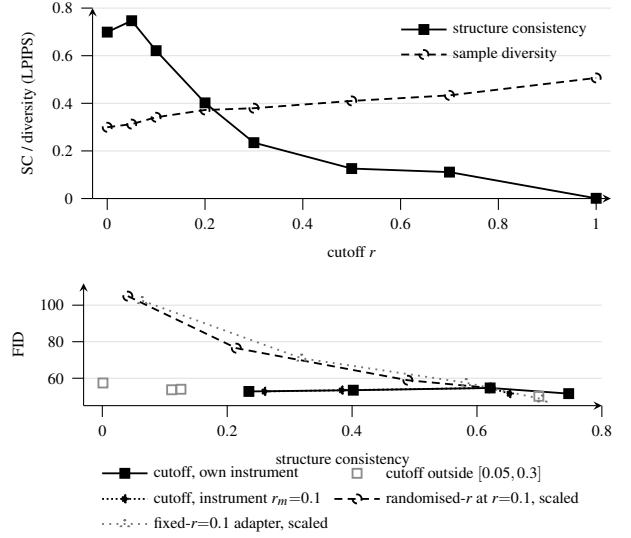


Figure 4: *Top:* structure consistency and sample diversity against the cutoff (values of Table 2, which gives SEM and seed sd). *Bottom:* FID against structure consistency, one FID estimate per point. Filled squares: the cutoff sweep scored at its own cutoff on $r \in [0.05, 0.3]$; open grey squares: the same sweep outside that range. Diamonds: the same generations re-scored with the instrument fixed at $r_m=0.1$. Circles and triangles: the randomised- r and fixed- $r=0.1$ adapters scaled at $r=0.1$, which already use $r_m=0.1$. On that one instrument the cutoff $r=0.3$ reaches 0.2599 at FID 52.77, while scale 0.50 reaches 0.2146 at FID 76.46. Both panels are plotted directly from the evaluation outputs.

4.5. Baselines

Against pretrained ControlNet [ZRA23] and T2I-Adapter-Canny [MWX*24], over the same test images and captions, each baseline is given Canny edges (thresholds 100 and 200) at conditioning scale 1.0, with the pipeline’s default PNDM sampler, 50 steps, CFG 7.5 and seed 2026; Table 4 lists them next to our adapter at $r=0.1$, which samples with DDIM. The ControlNet-HED checkpoint is fed the same Canny edges rather than HED boundaries. T2I-Adapter-Canny is a side network and so closer to our design than a copied encoder. Two asymmetries make this a comparison of operating points rather than a like-for-like control: the conditions differ (Canny edges against a frequency band), and our adapter was trained on 65,100 FFHQ images while ControlNet is zero-shot here and FID is measured against FFHQ statistics. On text alignment ControlNet-Canny’s CLIP 30.34 exceeds our best over the whole sweep, 30.30. We leave out T2I-Adapter’s sketch checkpoint, which expects PiDiNet sketches as input; fed Canny edges it would measure that mismatch rather than the method.

4.6. Outside the training domain

Everything above is FFHQ. To test whether the dial is specific to faces we ran the sweep on the 500 general-scene photographs of the public photograph-sketch set rhfeiyang/photo-sketch-pair-500 on Hugging Face, whose dataset card states neither a

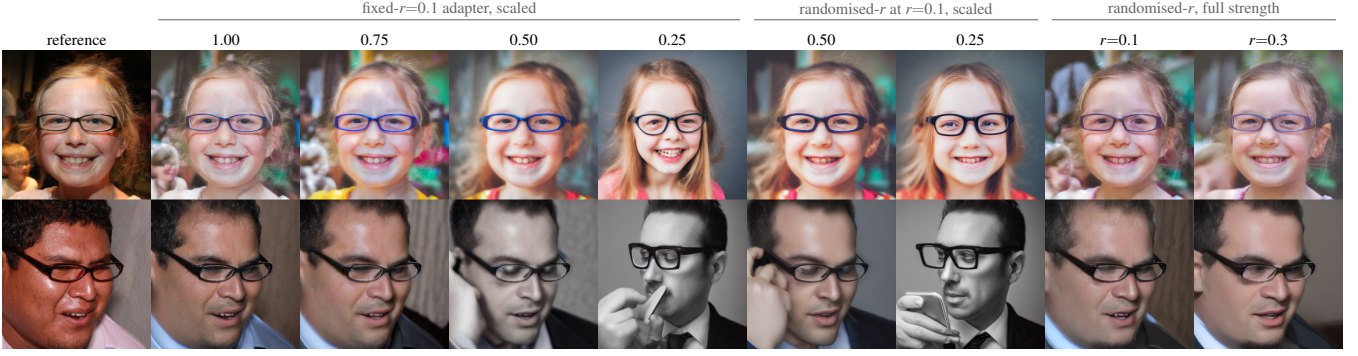


Figure 5: The first two test images under adapter strength and under the cutoff, sampling seed 2026 and BLIP caption throughout. Columns 2–5: the fixed- $r=0.1$ specialist at scales 1.00 to 0.25 (Table 3). Columns 6–7: the randomised- r adapter at $r=0.1$, scaled to 0.50 and 0.25. Columns 8–9: the randomised- r adapter at full strength at $r=0.1$ (Table 2) and $r=0.3$, the same generations as in Figure 3. At scale 0.75 the fixed- r adapter still follows the reference. Lowering the strength of either adapter to 0.50 changes background and colour, and in the second row the pose; at 0.25 the pose and framing change in both rows and the second row turns grayscale. Moving the cutoff to $r=0.3$ keeps the reference layout. All outputs are generated by the model under study.

r	SC $\pm$ SEM	seed sd	Edge-F1	Diversity	FID
0.00	0.6991 ± 0.0070	0.0105	0.7776	0.2994	50.13
0.05	0.7469 ± 0.0030	0.0015	0.7796	0.3126	51.58
0.10	0.6211 ± 0.0035	0.0059	0.7439	0.3414	54.68
0.20	0.4019 ± 0.0037	0.0098	0.7113	0.3719	53.45
0.30	0.2347 ± 0.0041	0.0106	0.6878	0.3794	52.77
$r > 0.3$					
0.50	0.1259 ± 0.0047	0.0093	0.6225	0.4100	53.97
0.70	0.1110 ± 0.0047	0.0186	0.5788	0.4334	53.69
1.00	0.0010 ± 0.0006	0.0002	0.4724	0.5066	57.40

Table 2: Operating curve, 500 of the 3,500 test images. All columns come from training seed 2026 except seed sd, which is the standard deviation over training seeds 2026–2028; SEM is over the 500 images. Edge-F1 is a cutoff-independent reading of the same runs and is reported alongside SC because SC re-extracts at the same cutoff the model was given (Sec. 4.1). FID has a single estimate per cell and no variance, so small differences between rows are not read as effects.

Setting	SC	Diversity	FID
<i>fixed-$r=0.1$ adapter, scaled</i>			
scale 0.25	0.0640	0.7689	102.43
scale 0.50	0.3193	0.5540	70.91
scale 0.75	0.5829	0.4044	57.52
scale 1.00	0.7050	0.2967	48.51
<i>randomised-r adapter at $r=0.1$, scaled</i>			
scale 0.25	0.0408	0.7752	104.95
scale 0.50	0.2146	0.6061	76.46
scale 0.75	0.4893	0.4171	58.90
<i>randomised-r adapter, cutoff moved</i>			
$r=0.2$	0.4019	0.3719	53.45
$r=0.3$	0.2347	0.3794	52.77
<i>same generations, instrument fixed at $r_m=0.1$</i>			
$r=0.05$	0.6524	0.3126	51.58
$r=0.2$	0.3838	0.3719	53.45
$r=0.3$	0.2599	0.3794	52.77

Table 3: Strength against cutoff. Every scaled row is generated and scored at $r=0.1$, so it already uses the instrument $r_m=0.1$; the cutoff-moved rows are scored at their own cutoff, and the last block re-scores the same generations with $r_m=0.1$ (Sec. 4.1). Scale 1.00 of the fixed- r adapter is a separate inference run of the specialist of Sec. 4.4 (0.7049 there).

licence nor an upstream source, centre-cropped to 512^2 , and, as an in-domain reference under the identical script, on the 500 FFHQ test images, both with the prompt “a high-quality photo”. Each generation is scored in two ways: the grayscale output against the grayscale input photograph, and the grayscale output against the filtered condition it was given. Neither filters the output, so neither is the structure-consistency instrument of Table 2, and the in-domain values differ from it (Table S7).

Against the input photograph the score falls from 0.4506 at $r=0.05$ to 0.1384 at $r=0.3$ out of domain, and from 0.5981 to 0.4142 in domain; against the filtered condition it falls from 0.3139 to 0.0303 out of domain and from 0.3193 to 0.0334 in domain. The dial carries out of domain. With the adapter off, the out-of-domain images score 0.0145 against the input photograph at $r=0.1$, against 0.3106 with it on.

Image quality does not transfer: FID, each run measured against its own input photographs, rises from 65.71 in domain to 148.97 out of it at $r=0.1$. Against the filtered condition, the same two runs score 0.1796 and 0.1830, and they stay this close at every cutoff. Both are far above chance: scored against a different input’s filtered condition the same generations reach 0.0015 and 0.0018 at $r=0.1$, and no reading in either run is fewer than 35 standard errors above its own floor. Condition following and image quality therefore come apart: on the range we claim, the dial holds structure

Method	Sampler	Params	Edge-F1	LPIPS	FID	CLIP
ControlNet-Canny	PNDM	361,279,120	0.6854	0.5294	71.92	30.34
ControlNet-HED, Canny input	PNDM	—	0.5733	0.6548	118.98	29.16
T2I-Adapter-Canny	PNDM	77,000,640	0.5845	0.5690	86.66	30.58
ours, $r=0.1$	DDIM	6,611,469	0.7439	0.3748	54.68	29.53

Table 4: Pretrained baselines given Canny edges against our adapter at $r=0.1$, on the same 500 test images and captions with 50 steps, CFG 7.5 and seed 2026. Params: trainable parameters for ControlNet-Canny and ours, side-network parameters for T2I-Adapter. The samplers differ, so the rows are operating points, not a like-for-like comparison.



Figure 6: The first four of the 40 face drawings in file order (top row), and the unedited zero-shot outputs of the main model for them (bottom row; prompt “a high-quality photo of a face”). The drawings carry no cutoff; the model has to read structure density off the drawing itself, which the per-image min-max normalisation permits. Most of the scans carry the archive filename stamped across the top, and the model redraws it — a literal reading of the condition it was given. Removing the whole top 20% of the frame (102 rows), a generous over-estimate of that band, moves the score to 0.5398 (-0.0086, 0.32 standard errors), so the band is not what the score is measuring.

about as closely outside the training domain as inside it, while FID does not transfer.

4.7. Hand-drawn sketches

Figure 6 shows the transfer. On all 40 real face drawings of the public Hugging Face set a573p/Face-Sketches40, which its card describes as images of the CUHK Face Sketch database and for which it states no licence. Every scan is stamped with a FERET filename, so they are from the FERET-based CUFSS set [WT09,ZWT11] rather than the photographic subsets of CUFSS. The drawings, which the model was never trained on, prompted with “a high-quality photo of a face”, the grayscale output correlates with the input drawing at 0.5484 and CLIP score is 23.43. FID is not reported: 40 images is too few for a stable estimate, and our evaluator refuses rather than print one. We also ran 500 general-scene drawings, the sketch half of the public photograph-sketch set rhfeiyang/photo-sketch-pair-500 on Hugging Face, far outside the training domain and with the same face prompt, where that correlation falls to 0.1714 and FID against

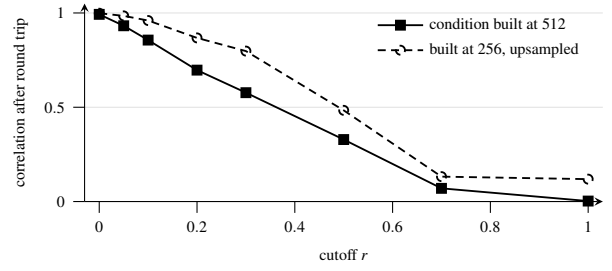


Figure 7: Correlation between the condition fed to the autoencoder and the same condition re-extracted after an encode-decode round trip, the first 24 FFHQ-512 images in file order, which are not the test split. Above $r \approx 0.5$ the condition is effectively gone. Building the condition at 256 and upsampling puts the structure at lower absolute frequency and preserves far more, which is why comparisons against older pipelines need the source resolution stated.

FFHQ is 285.91: the model follows the drawing but does not produce a face. It was trained on nothing else.

Sketch-based face editing has divided a drawing’s role before, taking geometry from the sketch and appearance from a reference [CLL*21], or splitting it into separately applicable sketch concepts [XF24]. The cutoff divides it by frequency and needs no reference image.

A drawing arrives without a cutoff, but the artist can give it one: filter the drawing at r and hand the model the result, the operation every training condition went through. On 40 face drawings, the grayscale output’s correlation with the original drawing falls from 0.5486 at $r=0$ to 0.3390 at $r=0.1$, 0.2271 at $r=0.3$ and 0.0407 at $r=1$; $r=0.5$ and $r=0.7$ lie within a standard error (0.0419) of each other. As intended use rather than measurement: $r=0.05$ is for holding the face as closely as the adapter can hold it, $r=0.3$ for keeping the layout while the detail is redrawn, and above that the layout itself is released. Changing r costs one regeneration, 1.88 s per image on one H200 (Table 1).

Against the filtered drawing the model was actually given, the same generations score 0.2552 at $r=0.05$, 0.1399 at $r=0.1$ and 0.0142 at $r=0.3$. Scored by the same script, 500 FFHQ test photographs prompted with “a high-quality photo” reach 0.3193, 0.1796 and 0.0334 (Sec. 4.6), so a filtered drawing is followed less closely than a filtered photograph at every cutoff; at $r=0.3$ the drawing reaches less than half the photograph’s score (0.0142 against 0.0334), though the prompts and set sizes differ (Table S8).

5. What the design needs

5.1. How much of the condition survives the frozen VAE

Figure 7 measures how much of a high-pass condition survives the frozen SD VAE. After an encode-decode round trip, the re-extracted condition correlates with the original at 0.8562 for $r=0.1$, 0.5773 for $r=0.3$ and 0.0700 for $r=0.7$. The round trip lower-bounds what the latent retains: a component may survive in the latent that the frozen decoder cannot render. The frozen-VAE designs in Table 5 inject through attention only and reach structure

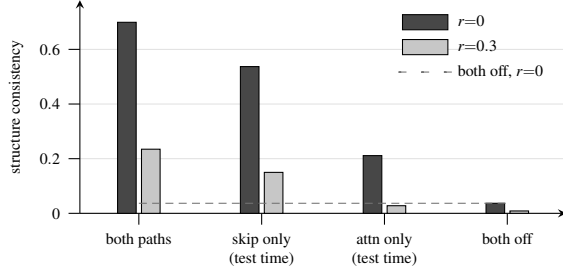


Figure 8: Both injection paths on, then each alone, then neither. The skip residuals carry most of the structure signal; cross-attention alone recovers little of it, and with both off the model falls to the frozen backbone (FID 120.50 against 50.13 with both paths on; skip only 65.12, attention only 67.07). The dashed line is the both-off control at $r=0$. Paths are switched off at test time on the trained main model; the separately trained skip-only model is in Table 7. At $r=0$ the single-path gains over the control roughly add up to the both-paths gain; at $r=0.3$ the both-paths gain over the control exceeds the sum of the single-path gains.

Cond. encoder	Injection	SC $r=0.05$	SC $r=0.1$	FID $r=0.1$
no adapter (control)		0.0074	0.0082	120.50
frozen VAE	attn, no zero-init gate	0.0234	0.0147	76.25
frozen VAE	attn	0.0303	0.0177	65.71
conv	attn	0.1485	0.0772	56.39
conv + 128-ch stem	attn	0.2826	0.1882	53.17
conv + 256-ch stem	attn	0.2986	0.2158	52.84
conv	attn, 7×7	0.0359	0.0218	58.51
conv	attn + skip	0.7469	0.6211	54.68

Table 5: Absolute configurations, not a cumulative chain. Rows below the control change one field of the conv / attn row; the first frozen-VAE row changes two (encoder, and no zero-initialised gate). All rows use a 3×3 window and a zero-initialised gate unless stated. The main model is the last row and carries no stem; its attention sites read the encoder output through a 1×1 projection. The frozen-VAE rows use attention injection only. They and the 7×7 row stay close to the no-adapter control at both cutoffs, so at that level a weak design and a run that failed to learn are not distinguishable. Stem width with the skip path present is varied in Table 6.

consistency 0.0177 at $r=0.1$, where the round trip still keeps a correlation of 0.8562, so condition loss in the round trip does not by itself account for their result. An image-prompt design such as IP-Adapter [YZL*23] reads a CLIP image embedding instead, so this measurement does not speak to it. Building the condition at 256 and upsampling moves the structure to lower absolute frequency, and the round trip then keeps 0.7978 at $r=0.3$ against 0.5773 at native resolution, a data-pipeline artefact worth knowing about when comparing against older setups.

5.2. Where to inject

Table 5 lists absolute configurations. Each row below the control changes one field of the conv / attn row, except the frozen-VAE row without a zero-initialised gate, which changes two. Figure 8 decomposes the main model by switching paths off at test time. Keeping only cross-attention drops SC at $r=0$ from 0.6991 to 0.2112; keeping only the skip path retains 0.5370. With both paths off the model falls to 0.0368 SC and FID 120.50, the control that says the adapter is doing the work. The two single paths are close on FID (65.12 against 67.07) and far apart on SC. How the two paths combine also changes with the cutoff: at $r=0$ their gains over the control roughly add up, while at $r=0.3$ the two together exceed that sum. We read this as the paths carrying the same information when the condition is dense and different information when it is sparse, the skip residuals placing local detail and cross-attention saying where it goes. We did not measure this.

Within the attention-only design, a 128-channel stem raises SC at $r=0.05$ from 0.1485 to 0.2826, and a 256-channel stem reaches 0.2986 with 6,998,637 trainable parameters against 405,613 without a stem; the skip path trained alone, with 6,205,856 parameters, reaches 0.7085 (Sec. 5.3). A 7×7 window gives 0.0359, far below the 3×3 design (0.1485) and close to the no-adapter control (0.0074). All of these vary capacity with the skip path absent, so they measure capacity within the attention-only design.

Table 6 repeats the one-field ablations on the main model with both paths on. Each is a single seed, so we read it directionally against the main model’s three-seed spread at each cutoff (0.0105 at $r=0$, 0.0015 at $r=0.05$). Capacity helps here too. At $r=0$ a 256-channel stem lifts SC from 0.6991 to 0.7900, several times the seed spread there; adding the skip path to attention alone moves it from 0.2112 to 0.6991. At $r=0.05$ the stem gain shrinks (0.7469 to 0.7631), and FID at $r=0.1$ drops from 54.68 to 51.36. The attention window does not help: 1×1 matches 3×3 within the seed spread or exceeds it at every cutoff listed, and 5×5 is lower at all of them, so what the attention sites contribute is where the condition enters, not a neighbourhood. The zero-initialised gate is worth keeping; without it SC is lower from $r=0$ to 0.3 and FID at $r=0.1$ is higher. The largest single change is how r is drawn. A log-uniform sampler lifts SC at $r=0$ to 0.9480 and lowers FID there to 42.06, but gives up the upper range (0.0858 at $r=0.5$ against 0.1259); it moves the dial’s resolution toward small r rather than improving it everywhere. The results above use the uniform sampler. The main model keeps the 3×3 window and no stem, although the 1×1 and stem variants score higher at several cutoffs in Table 6, because these one-field ablations were trained after the main configuration and its three seeds had been fixed. The same holds for the $r=u^2$ sampler. It concentrates training at small r and scores higher structure consistency than the main model at each measured cutoff of the claimed range (Tables 6 and S1). Its FID is lower at $r=0.1$ and within single-estimate noise at $r=0.05$ and 0.3 (Table S2). We leave it as an ablation rather than adopt it: the sampler is one of the fields this section varies, and choosing it on these numbers would be choosing it on the test set.

5.3. The skip path alone

Section 5.2 ablates the two injection paths on a model trained with both. That confounds placement with capacity: the cross-attention branch is still present and trained, just switched off at test time.

One field changed	SC				FID
	$r=0$	0.05	0.3	0.5	$r=0.1$
main model	0.6991	0.7469	0.2347	0.1259	54.68
seed sd (3 seeds)	0.0105	0.0015	0.0106	0.0093	—
1×1 window	0.7825	0.7640	0.2271	0.1255	55.70
5×5 window	0.6764	0.7175	0.2039	0.0944	55.68
no zero-init gate	0.6838	0.7322	0.2058	0.1290	57.02
average-pool downsampling	0.7203	0.7465	0.2279	0.1241	55.13
64-ch stem	0.7775	0.7594	0.2354	0.1170	51.91
256-ch stem	0.7900	0.7631	0.2511	0.1214	51.36
colour condition	0.6949	0.7463	0.2343	0.1255	54.75
no condition dropout	0.7042	0.7353	0.2279	0.1231	54.69
$r=u^2$ sampler	0.8936	0.7836	0.2468	0.1039	53.46
log-uniform sampler	0.9480	0.8000	0.2302	0.0858	54.87

Table 6: One-field ablations on the main model, skip and attention paths both on, one training seed each; the second row is the main model’s spread over three seeds. The main model uses a 3×3 window, a zero-initialised gate, learned 2×2 strided downsampling of the condition features at each attention site, no stem before the 1×1 projection, a grayscale condition, condition dropout 0.05, and r uniform on $[0, 1]$. Drawing $r = u^2$ with u uniform concentrates training at small r ; the log-uniform sampler on $[10^{-3}, 1]$ does so more strongly.

r	skip only	both paths	attention only
0.05	0.7085	0.7469	—
0.10	0.5765	0.6211	0.0275
0.20	0.3508	0.4019	—
0.30	0.1608	0.2347	0.0279
0.50	0.0709	0.1259	—
0.70	0.0757	0.1110	—

Table 7: Structure consistency for a model trained with the skip path only, against the main model and against its cross-attention branch measured in isolation. The last column comes from switching a path off at test time, so it is not a trained-from-scratch comparison.

We therefore trained the skip path from scratch, with the attention branch removed entirely.

It reaches SC 0.5765 at $r=0.1$ against 0.6211 for both paths and 0.0275 for cross-attention alone, at FID 58.20 against 54.68. Trained on its own, the skip path recovers most but not all of the main model, so the attention branch adds something the skip path does not supply. The skip-only model trains 6,205,856 parameters against 6,611,469 for both paths, and the attention-only design with a 256-channel stem, at 6,998,637 parameters, reaches 0.2158 at $r=0.1$ (Table 5). The full curve is in Table 7.

6. Limitations

Everything here is one domain (FFHQ faces) and one backbone (SD1.5), and the three results depend on that backbone to different degrees. The dial is defined on the condition alone; we measured it on SD1.5 with FFHQ-512. SDXL [PEL*24] keeps the UNet

skip tensors and is the closest next test. The skip-path injection is specific to UNets: MMDiT backbones [EKB*24] have no UNet skip connections, and control there enters through shared tokens or LoRA [HSW*22] modules [TLY*25, ZYS*25, LZL*25]. The VAE measurement covers the four-channel SD autoencoder.

We claim the dial on $r \in [0.05, 0.3]$. Above that range the condition retains little energy (0.0074 of the non-constant spectral energy at $r=0.5$), and structure consistency becomes hard to interpret, since two nearly empty signals correlate near zero whatever the model does. FID is a single estimate per cell with no variance, so we leave gaps inside the band of the cutoff sweep (50.13–57.40) uninterpreted. The strength setting’s FID is measured against FFHQ statistics; at scales of 0.50 and below the samples of Figure 5 depart from the reference layout rather than visibly degrading, so the rise reflects leaving that distribution at least in part, and we report no no-reference quality measure. The one-field ablations of Table 6 are single-seed, and the per-image normalisation of the condition was fixed by design and never ablated. At $r=0.1$ the fixed-cutoff specialist leads the randomised- r adapter by +13.55 seed sd (Sec. 4.4). The frozen VAE was never paired with the skip path, so in Table 5 the VAE result is confounded with the weakness of attention-only injection. The sketch results have no adapter-off control. The ControlNet comparison is between operating points; a ControlNet trained on our condition and data would be the like-for-like control, and we did not train one. We do not compare against a sketch-reliability control [YWLG20] or against noise-level editing [MHS*22], training-free filter guidance [GYD24] or geometry–appearance face editing [CLL*21]; they act on different inputs and would need their own protocol.

The adapter trains 6,611,469 parameters, fewer than ControlNet and more than recent lightweight control modules: SimpleControlNet [FCS*25] reports under 3M parameters on SD1.5, ControlNet-XS [ZFR24] ships much smaller control networks than ControlNet, and the DiT modules above use a fraction of our count. We make no efficiency claim. We evaluate the dial with image metrics only, and neither SC nor Edge-F1 separates layout from appearance, so the observation that layout holds while appearance changes rests on the samples in Figures 3 and 5. Whether equal steps in r are perceived as equal changes, and whether artists find the control useful, is not measured here.

7. Conclusion

Randomising the bandwidth of a structure condition during training turns the cutoff into an inference-time control, and that control is not reproducible by scaling this adapter’s strength. One checkpoint covers $r \in [0.05, 0.3]$ at FID between 51.58 and 54.68; across that range our samples keep the face layout while background, clothing and colour follow the reference less. Lowering strength to a comparable looseness costs FID (Sec. 4.3). The checkpoint trails a specialist at $r=0.1$ and leads one at $r=0.3$ (Sec. 4.4). The design reads the condition with a convolutional encoder rather than the frozen autoencoder and injects it through the UNet skip tensors and cross-attention, both choices measured on SD1.5.

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